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### Vehicle having an energy absorption member

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#### Abstract

A vehicle may include a vehicle body including a floor panel, a battery pack located below the floor panel, and an energy absorption member located outwardly from the battery pack in a vehicle width direction and fixing the battery pack to the vehicle body. The energy absorption member may be a hollow member extending along a vehicle longitudinal direction and having a constant cross section along the vehicle longitudinal direction. An inner sidewall of the energy absorption member that faces the battery pack may incline downwardly in a vehicle height direction as it extends inwardly in the vehicle width direction.

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## Background/Summary

### CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application claims priority to Japanese patent application No. 2022-053201, filed on Mar. 29, 2022, the entire contents of which are incorporated herein by reference.

### TECHNICAL FIELD

(2) The technique disclosed herein relates to a vehicle.

### BACKGROUND

(3) Japanese Patent Application Publication No. 2017-226353 describes a vehicle. This vehicle includes a pair of rockers, a floor panel extending between the pair of rockers, a battery pack located below the floor panel, and an energy absorption member located outwardly from the battery pack in a vehicle width direction and fixed to one of the pair of rockers.

## SUMMARY

- (4) With the above structure, when side collision occurs to the vehicle, the energy absorption member (which hereafter may be referred to as an EA member) deforms, by which collision energy is absorbed. At this time, if an inner sidewall of the EA member facing the battery pack is pressed against a sidewall of the battery pack, there is a risk that relatively large collision load may be applied on the sidewall of the battery pack.
- (5) In light of the above situation, the present disclosure provides a technique which may reduce collision load transmitted from an EA member to a sidewall of a battery pack.
- (6) The technique disclosed herein is implemented as a vehicle. This vehicle may comprise: a vehicle body comprising a floor panel; a battery pack located below the floor panel; and an energy absorption member located outwardly from the battery pack in a vehicle width direction and fixing the battery pack to the vehicle body. The energy absorption member may be a hollow member extending along a vehicle longitudinal direction and having a constant cross section along the vehicle longitudinal direction and comprises an inner sidewall that faces the battery pack. The inner sidewall of the energy absorption member may incline downwardly in a vehicle height direction as it extends inwardly in the vehicle width direction.
- (7) With the above structure, when side collision occurs to the vehicle, the EA member deforms, by which collision energy is absorbed. Here, the inner sidewall of the EA member facing the battery pack inclines downwardly in the vehicle height direction as it extends inwardly in the vehicle width direction. Thus, even when the EA member is compressed and deformed in the vehicle width direction toward the battery pack, it is possible to avoid or suppress the EA member coming into contact with the sidewall of the battery pack. Consequently, collision load applied on the sidewall of the battery pack from the EA member can be reduced.
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## Description

### BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 schematically illustrates a configuration of a vehicle **10** of Embodiment 1.
- (2) FIG. 2 is a view explaining positional relationships between a battery pack **20**, a pair of EA members **30**, **32** and a pair of rockers **24**, **26**.
- (3) FIG. 3 is a cross-sectional view along a line of FIG. 2.
- (4) FIG. 4A illustrates an example of deformation of the EA member **30** over time when side collision occurs to the vehicle **10**.
- (5) FIG. 4B illustrates the example of deformation of the EA member **30** over another time when side collision occurs to the vehicle **10**.
- (6) FIG. 4C illustrates the example of deformation of the EA member **30** over another time when side collision occurs to the vehicle **10**.
- (7) FIG. 5 is an enlarged view of an enclosed area V in FIG. 3.
- (8) FIG. 6 is a cross-sectional view illustrating a structure of a vehicle **10** of Embodiment 2 at a position corresponding to the line in FIG. 2.

### DETAILED DESCRIPTION

- (9) In an embodiment of the technique disclosed herein, the vehicle may further comprise a collar extending in the vehicle height direction and comprising an upper end being in contact with the battery pack and a lower end being in contact with the energy absorption member; and a bolt extending through the collar and fastening the energy absorption member to the battery pack. According to such a configuration, a distance between the battery pack and the EA member fastened by the bolt is determined by a dimension of the collar in the vehicle height direction. Thus, variations in the above distance due to a fastening operation can be avoided or suppressed.
- (10) In the aforementioned embodiment, the inner sidewall of the energy absorption member may

comprise an opening through which the collar extends. According to such a configuration, the collar can be disposed relative to the energy absorbing member regardless of an angle at which the sidewall inclines.

(11) In some of the aforementioned embodiments, the collar may be welded to the inner sidewall of the energy absorption member. According to such a configuration, the collar can tightly be fixed to the EA member. Thus, even when the EA member is compressed and deformed in the vehicle width direction toward the battery pack, it is possible to avoid or suppress both the EA member and the collar coming into contact with the sidewall of the battery pack.

(12) In some of the aforementioned embodiments, an inner diameter of the collar is at least twice a nominal diameter of the bolt. According to such a configuration, a relatively large space can be secured between the collar and the bolt. Thus, when the EA member is compressed and deformed in the vehicle width direction toward the battery pack, the space between the collar and the bolt can reduce collision load applied from the EA member on the sidewall of the battery pack.

(13) In some of the aforementioned embodiments, the battery pack may comprise a bracket that protrudes outwardly in the vehicle width direction. In this case, the bolt may fasten the energy absorption member to the bracket of the battery pack. According to such a configuration, a distance between the battery pack and the EA member can be increased in the vehicle width direction by an amount by which the bracket protrudes outwardly in the vehicle width direction.

(14) Thus, when the EA member is compressed and deformed in the vehicle width direction toward the battery pack, it is possible to reduce the collision load applied from the EA member on the sidewall of the battery pack. In another embodiment, however, the EA member may directly be fastened to the battery pack by the bolt without disposing the bracket on the battery pack.

(15) In an embodiment of the technique disclosed herein, the energy absorption member may comprise an inner end portion in which the inner sidewall is included and a middle portion adjacent to the inner end portion in the vehicle width direction. In this case, a thickness of the inner sidewall of the energy absorption member may be greater than a thickness of any one of walls constituting the middle portion of the energy absorption member. According to such a configuration, it is possible to make rigidity of the inner sidewall of the EA member facing the battery pack greater than rigidity of any one of the walls constituting the middle portion of the EA member. Thus, when the EA member is compressed and deformed in the vehicle width direction toward the battery pack, it is possible to make the inner sidewall of the EA member less susceptible to deformation as compared to any one of the walls constituting the middle portion of the EA member. Consequently, it is possible to avoid or suppress the EA member coming into contact with the side wall of the battery pack.

(16) In the aforementioned embodiment, a thickness of any one of walls constituting the inner end portion may be greater than the thickness of any one of the walls constituting the middle portion of the energy absorption member. According to such a configuration, it is possible to make rigidity of any one of the walls constituting the inner end portion including the inner sidewall facing the battery pack greater than the rigidity of any one of the walls constituting the middle portion. Thus, when the EA member is compressed and deformed in the vehicle width direction toward the battery pack, it is possible to make the inner end portion of the EA member less susceptible to deformation as compared to the middle portion.

(17) In an embodiment of the technique disclosed herein, the battery pack may comprise a bracket that protrudes outwardly in the vehicle width direction. The energy absorption member may comprise an inner end portion in which the inner sidewall is included and a middle portion adjacent to the inner end portion in the vehicle width direction. The inner end portion of the energy absorption member may be fixed to a lower surface of the bracket of the battery pack from below. In this case, an upper wall of the middle portion of the energy absorption member may be located lower than the lower surface of the bracket in the vehicle height direction. According to such a configuration, when the EA member is compressed and deformed in the vehicle width direction

toward the battery pack, it is possible to avoid or suppress the EA member coming into contact with the bracket of the battery pack. Consequently, the collision load applied on the battery pack from the EA member can be reduced.

(18) Simple recitations such as “forward”, “rearward”, “a longitudinal direction” herein respectively refer to forward, rearward and a longitudinal direction with respect to a vehicle. Similarly, simple recitations such as “leftward”, “rightward” and “a vehicle width direction” respectively refer to leftward, rightward and a vehicle width direction with respect to the vehicle and simple recitations such as “upward”, “downward” and “a height direction” refer to upward, downward and a height direction with respect to the vehicle. The vehicle width direction of the vehicle is also a left-right direction of the vehicle, and it may be referred to as “left-right direction” herein. For example, when the vehicle is placed on a horizontal surface, the height direction of the vehicle coincides with the vertical direction. The vehicle width direction of the vehicle is a direction parallel to the horizontal surface and parallel to an axle of the vehicle, and the longitudinal direction of the vehicle is a direction parallel to the horizontal surface and perpendicular to the axle of the vehicle.

(19) Representative, non-limiting examples of the present disclosure will now be described in further detail with reference to the attached drawings. This detailed description is merely intended to teach a person of skill in the art further details for practicing aspects of the present teachings and is not intended to limit the scope of the present disclosure. Furthermore, each of the additional features and teachings disclosed below may be utilized separately or in conjunction with other features and teachings to provide improved vehicles, as well as methods for using and manufacturing the same.

(20) Moreover, combinations of features and steps disclosed in the following detailed description may not be necessary to practice the present disclosure in the broadest sense, and are instead taught merely to particularly describe representative examples of the present disclosure. Furthermore, various features of the above-described and below-described representative examples, as well as the various independent and dependent claims, may be combined in ways that are not specifically and explicitly enumerated in order to provide additional useful embodiments of the present teachings.

(21) All features disclosed in the description and/or the claims are intended to be disclosed separately and independently from each other for the purpose of original written disclosure, as well as for the purpose of restricting the claimed subject matter, independent of the compositions of the features in the embodiments and/or the claims. In addition, all value ranges or indications of groups of entities are intended to disclose every possible intermediate value or intermediate entity for the purpose of original written disclosure, as well as for the purpose of restricting the claimed subject matter.

(22) (Embodiment 1) With reference to drawings, a vehicle **10** of Embodiment 1 will be described. The vehicle **10** of the present embodiment is classified as an electric vehicle including a motor **16** driving wheels **14f**, **14r**, and is typically an electric vehicle which travels on roads (a so-called automobile). However, a part of or all the technique described in the present embodiment can similarly be applied to an electric vehicle which travels on tracks. The vehicle is not limited to a vehicle operated and driven by a user, but may be a vehicle remotely operated by external device(s) or a vehicle which travels autonomously.

(23) Here, a direction FR in the drawings indicates a front side in a longitudinal direction (or a front-rear direction) of the vehicle **10**, and a direction RR indicates a rear side in the longitudinal direction of the vehicle **10**. A direction LH indicates a left side in a vehicle width direction (or a left-right direction) of the vehicle **10** and a direction RH indicates a right side in the vehicle width direction of the vehicle **10**. A direction UP indicates an upside in a height direction (or an up-down direction) of the vehicle **10**, and a direction DW indicates a downside in the height direction of the vehicle **10**.

(24) As illustrated in FIG. 1, the vehicle **10** includes a vehicle body **12** and the plurality of wheels **14f**, **14r**. The vehicle body **12** includes a cabin **12c** which is a space to accommodate occupants. The plurality of wheels **14f**, **14r** is rotatably attached to the vehicle body **12**. The plurality of wheels **14f**, **14r** includes a pair of front wheels **14f** positioned at a front portion of the vehicle body **12** and a pair of rear wheels **14r** positioned at a rear portion of the vehicle body **12**. The pair of front wheels **14f** is disposed coaxially to each other, and the pair of rear wheels **14r** is also disposed coaxially to each other. The number of wheels **14f**, **14r** is not limited to four. The vehicle body **12** is constituted of, but not particularly limited to, metal, such as steel or aluminum alloy.

(25) As illustrated in FIG. 1, the vehicle **10** further includes a motor **16**, an electric control unit **18** and a battery pack **20**. The motor **16** is a traction motor that drives the pair of rear wheels **14r** and is connected to the pair of rear wheels **14r**. The battery pack **20** is an electric device configured to supply power to the motor **16** and is electrically connected to the motor **16** via the electric control unit **18**. The battery pack **20** houses a plurality of secondary battery cells and is configured rechargeable with external power or regenerative power of the motor **16**. The battery pack **20** is located below a floor panel **22** and is disposed along the floor panel **22**. Although this is merely an example, as illustrated in FIGS. 2 and 3, the battery pack **20** of the present embodiment includes brackets **20a** protruding outwardly in the vehicle width direction. Moreover, although not particularly limited to this configuration, a support **20b** protruding downwardly is disposed on a lower wall of the battery pack **20**. The electric control unit **18** houses a DC-DC converter and/or inverter and is configured to control driving power supplied from the battery pack **20** to the motor **16** and regenerative power supplied from the motor **16** to the battery pack **20** depending on, for example, driving operations by the user.

(26) Wheel(s) to be driven by the motor **16** are not limited to the pair of rear wheels **14r**, and the motor **16** simply needs to be configured to drive at least one of the plurality of wheels **14f**, **14r**. The vehicle **10** may further include another prime mover such as an engine instead of or as an alternative to the motor **16**. The vehicle **10** may include another power device such as fuel cell unit(s) or solar panel(s) instead of or as an alternative to the battery pack **20**. The vehicle **10** is not limited to an electric automobile described herein but may be a hybrid vehicle, fuel-cell vehicle, a solar car, or the like.

(27) As illustrated in FIGS. 2 and 3, the vehicle body **12** includes the floor panel **22**, a pair of rockers **24**, **26** and a floor crossbeam **28**. The floor panel **22** is a plate-like member constituting a bottom surface of the cabin **12c**. Each of the rockers **24**, **26** is a hollow member having a constant cross-section along the vehicle longitudinal direction, and constitutes a frame of the vehicle body **12**. The pair of rockers **24**, **26** includes a left rocker **24** and a right rocker **26**. The left rocker **24** is positioned at a left edge **22a** of the floor panel **22** and extends in the front-rear direction on the outer side of the floor panel **22** in the vehicle width direction. The pair of rockers **24** is arranged symmetrically to each other with respect to the vehicle width direction. Thus, the right rocker **26** is positioned at a right edge of the floor panel **22** and extends in the front-rear direction on the outer side of the floor panel **22** in the vehicle width direction.

(28) Although not particularly limited to this configuration, the left rocker **24** includes a left rocker inner panel **24a** positioned on the inner side in the width direction and a left rocker outer panel **24b** positioned on the outer side in the width direction. The left rocker inner panel **24a** and the left rocker outer panel **24b** are joined to each other at the respective top and bottom edges, and a closed space extending in the front-rear direction is defined inside the left rocker **24**. Although not particularly limited to this configuration as well, the right rocker **26** includes a right rocker inner panel **26a** positioned on the inner side in the width direction and a right rocker outer panel **26b** positioned on the outer side in the width direction. The right rocker inner panel **26a** and the right rocker outer panel **26b** are joined to each other at the respective top and bottom edges, and a closed space extending in the front-rear direction is defined inside the right rocker **26**. The floor panel **22** extends between the pair of rockers **24**, **26** and is joined to the left rocker inner panel **24a** at its left

edge **22a** and joined to the right rocker inner panel **26a** at its right edge. The rockers **24**, **26** may be constituted of three or more panels, not limited only to the rocker inner panels **24a**, **26a** and the rocker outer panels **24b**, **26b**.

(29) The floor crossbeam **28** is a hollow member having a constant cross section along the vehicle longitudinal direction and constitutes a part of the frame of the vehicle body **12**. The floor crossbeam **28** is positioned on the floor panel **22** and extends between the pair of rockers **24**, **26** in the vehicle width direction. Although the illustration is omitted, a plurality of floor crossbeams **28** is disposed between the pair of rockers **24**, **26**.

(30) As illustrated in FIGS. **2**, **3**, the vehicle **10** further includes a pair of energy absorption members **30**, **32** (hereafter referred to as EA members **30**, **32**). Each of the pair of EA members **30**, **32** is a hollow member extending along the vehicle longitudinal direction and having a constant cross section along the vehicle longitudinal direction. The pair of EA members **30**, **32** includes a left EA member **30** and a right EA member **32**. The pair of EA members **30**, **32** is arranged symmetrically to each other with respect to the vehicle width direction. The left EA member **30** is located at a left portion of the vehicle **10** and outwardly from the battery pack **20** in the vehicle width direction. The right EA member **32** is located at a right portion of the vehicle **10** and outwardly from the battery pack **20** in the vehicle width direction. At least one of the EA members **30**, **32** is compressed and deformed when side collision occurs to the vehicle **10**, by which it absorbs collision energy. The pair of EA members **30**, **32** is constituted of metal such as aluminum. However, a material constituting the pair of EA members **30**, **32** is not particularly limited.

(31) As described above, the pair of EA members **30**, **32** is arranged symmetrically to each other with respect to the vehicle width direction. Structures of the left EA member **30** and its relevant members (e.g., the battery pack **20** and the left rocker **24**) are symmetrical to structures of the right EA member **32** and its relevant members (e.g., the battery pack **20** and the right rocker **26**) with respect to the vehicle width direction. Thus, with reference to FIG. **3**, the structures of the left EA member **30** and its relevant members (e.g., the battery pack **20** and the left rocker **24**) will hereafter be described. Hereafter, the left EA member **30** will simply be referred to as the EA member **30**.

(32) As illustrated in FIG. **3**, the EA member **30** includes an inner end portion **34**, a first middle portion **36**, a second middle portion **38** and an outer end portion **40**. The inner end portion **34** is at the innermost position in the vehicle width direction and faces the battery pack **20**. The outer end portion **40** is at the outermost position in the vehicle width direction. The first middle portion **36** and the second middle portion **38** are positioned between the inner end portion **34** and the outer end portion **40** in the vehicle width direction. The first middle portion **36** is adjacent to the inner end portion **34** and the second middle portion **38** is adjacent to the outer end portion **40**. In other words, the inner end portion **34**, the first middle portion **36**, the second middle portion **38** and the outer end portion **40** are arranged in this order from inside to outside in the vehicle width direction.

(33) The inner end portion **34** includes an inner sidewall **34a**, a vertical wall **34b**, and a lower wall **34c**. The inner sidewall **34a** faces the battery pack **20** and inclines downwardly in the vehicle height direction as it extends inwardly in the vehicle width direction. The vertical wall **34b** extends along the vehicle height direction and the lower wall **34c** extends along the vehicle width direction. Thus, a space having a constant cross-section in the vehicle longitudinal direction is defined by the inner sidewall **34a**, the vertical wall **34b** and the lower wall **34c**.

(34) Although not particularly limited to this configuration, the EA member **30** further includes a protrusion **35**. The protrusion **35** is connected to the lower wall **34c** and the inner sidewall **34a** of the inner end portion **34** and extends toward the battery pack **20**.

(35) As illustrated in FIG. **3**, the vehicle **10** further includes first collars **42** and first bolts **44**. The inner sidewall **34a** included in the inner end portion **34** of the EA member **30** comprises openings through which the first collars **42** extend. The first collars **42** are disposed such that the first collars **42** extend through the openings in the inner sidewall **34a** in the vehicle height direction. Although this is an example, the first collars **42** are welded to the inner sidewall **34a** of the EA member **30**.

The upper end of each first collar **42** is in contact with corresponding one of the brackets **20a** of the battery pack **20** and the lower end of each first collar **42** is in contact with the lower wall **34c** of the inner end portion **34**. The first bolts **44** extend through the first collars **42** and fasten the EA member **30** to the brackets **20a** of the battery pack **20**. Thus, the EA member is fixed to the brackets **20a** of the battery pack **20** at the inner end portion **34**. An inner diameter **D1** of the first collars **42** is greater than a nominal diameter **D2** of the first bolts **44**. For example, the inner diameter **D1** of the first collars **42** is at least twice the nominal diameter **D2** of the first bolts **44**. As another embodiment, the brackets **20a** may not be disposed on the battery pack **20**, and the EA member **30** may directly be fastened to the battery pack **20** by the first bolts **44**.

(36) As illustrated in FIG. 3, the vehicle **10** further includes second collars **46** and second bolts **48**. Although not particularly limited to this configuration, an upper wall **38a** of the second middle portion **38** of the EA member **30** also includes openings through which the second collars **46** extend. The second collars **46** are disposed such that the second collars **46** extend through the openings in the upper wall **38a** of the second middle portion **38** in the vehicle height direction. Although this is an example, the second collars **46** are welded to the upper wall **38a** of the second middle portion **38**. The upper ends of the second collars **46** are in contact with the left rocker inner panel **24a** of the left rocker **24** and the lower ends of the second collars **46** are in contact with the lower wall **38b** of the second middle portion **38**. The second bolts **48** extend through the second collars **46** and fasten the EA member **30** to the left rocker **24**. Thus, the EA member **30** is fixed to the left rocker **24** of the vehicle body **12** at the second middle portion **38**. As described above, the EA member **30** is fixed to the brackets **20a** of the battery pack **20** at the inner end portion **34**, thus the EA member **30** fixes the battery pack **20** to the vehicle body **12**. Although not particularly limited to this configuration, the inner diameter of the second collars **46** is greater than the nominal diameter of the second bolts **48**.

(37) When side collision occurs to the vehicle **10**, in the structure described above, one of the EA members **30**, **32** deforms as illustrated in FIGS. 4A to 4C by which collision energy is absorbed. Here, the inner sidewall **34a** of each of the EA members **30**, **32** facing the battery pack **20** inclines downwardly in the vehicle height direction as it extends inwardly in the vehicle width direction. Thus, even when one of the EA members **30**, **32** is compressed and deformed in the vehicle width direction toward the battery pack **20**, it is possible to avoid or suppress the one of the EA members **30**, **32** coming into contact with the corresponding sidewall of the battery pack **20**. Consequently, collision load applied on the sidewall of the battery pack **20** from the one of the EA members **30**, **32** can be reduced.

(38) Although not particularly limited to this configuration, as illustrated in FIG. 5, a thickness **T1** of the inner sidewall **34a** is greater than each of thicknesses **T4**, **T5** of walls constituting the first middle portion **36**. According to such a configuration, it is possible to make rigidity of the inner sidewall **34a** of each of the EA members **30**, **32** facing the battery pack **20** greater than rigidity of any one of the walls constituting the first middle portion **36** of each of the EA members **30**, **32**. Consequently, when one of the EA members **30**, **32** is compressed and deformed in the vehicle width direction toward the battery pack **20**, it is possible to make the corresponding inner sidewall **34a** of the one of the EA members **30**, **32** less susceptible to deformation as compared to any one of the walls constituting the first middle portion **36** of the one of the EA members **30**, **32**. Thus, it is possible to avoid or suppress the one of the EA members **30**, **32** coming into contact with the corresponding sidewall of the battery pack **20**.

(39) Although not particularly limited to this configuration, as illustrated in FIG. 5, in addition to the thickness **T1** of the inner sidewall **34a**, a thickness **T2** of the vertical wall **34b** and a thickness **T3** of the lower wall **34c** is each greater than each of the thicknesses **T4**, **T5** of the walls constituting the first middle portion **36**. In other words, each of all the thicknesses **T1** to **T3** of the walls **34a** to **34c** constituting the inner end portion **34** is greater than each of the thicknesses **T4**, **T5** of the walls constituting the first middle portion **36**. According to such a configuration, it is



possible to make rigidity of each of the walls **34a** to **34c** of the inner end portion **34** including the inner sidewall **34a** facing the battery pack **20** greater than rigidity of any one of the walls constituting the first middle portion **36**. Consequently, when one of the EA members **30, 32** is compressed and deformed in the vehicle width direction toward the battery pack **20**, it is possible to make the corresponding inner end portion **34** of the one of the EA members **30, 32** less susceptible to deformation than the corresponding middle portion **36**.

(40) In another embodiment, however, the thickness **T2** of the vertical wall **34b** and the thickness **T3** of the lower wall **34c** may each be equal to the thickness **T4** (and/or **T5**) of each wall constituting the first middle portion **36** and only the thickness **T1** of the inner sidewall **34a** may be greater than the thickness **T4** (and/or **T5**) of each wall constituting the first middle portion **36**. Alternatively, in yet another embodiment, the thickness **T1** of the inner sidewall **34a**, the thickness **T2** of the vertical wall **34b** and the thickness **T3** of the lower wall **34c** may each be equal to the thickness **T4** (and/or **T5**) of each wall constituting the first middle portion **36**.

(41) (Embodiment 2) Next, with reference to FIG. 6, a vehicle of Embodiment 2 will be described. As compared to the vehicle **10** of Embodiment 1, the upper wall **36a** of the first middle portion **36** (more specifically, the upper surface of the upper wall **36a**) of each of the EA members **30, 32** of the vehicle of the present embodiment inclines downwardly as it extends outwardly in the vehicle width direction instead of the upper wall **36a** of the first middle portion **36** of each of the EA members **30, 32** being at an equal height even as it extends outwardly in the vehicle width direction. The inner end portion **34** of each of the EA members **30, 32** is attached to the lower surfaces **20c** of the corresponding brackets **20a** of the battery pack **20** from below. The upper wall **36a** of the first middle portion **36** of each of the EA members **30, 32** is located lower than the lower surfaces **20c** of the corresponding brackets **20a** in the vehicle height direction. According to such a configuration, when one of the EA members **30, 32** is compressed and deformed in the vehicle width direction toward the battery pack **20**, it is possible to avoid or suppress the one of the EA members **30, 32** coming into contact with the sidewalls of the corresponding brackets **20a** of the battery pack **20**. Consequently, collision load applied from the one of the EA members **30, 32** on the battery pack **20** can be reduced. It should be noted that explanations overlapping between Embodiment 1 and the present embodiment are omitted by giving the same reference numbers to the common configurations.

(42) The configuration of the present embodiment can also be employed in Embodiment 1. In other words, even when the upper wall **36a** of the first middle portion **36** of each of the EA members **30, 32** is at the same height even as it extends outwardly in the vehicle width direction, the inner end portion **34** of each of the EA members **30, 32** is attached to the lower surfaces **20c** of the corresponding brackets **20a** of the battery pack **20** from below and it is possible to position the upper wall **36a** of the first middle portion **36** of each of the EA members **30, 32** lower than the lower surfaces **20c** of the corresponding brackets **20a** in the vehicle height direction. According to such a configuration as well, when one of the EA members **30, 32** is compressed and deformed, collision load applied from the one of the EA members **30, 32** on the corresponding sidewall of the battery pack **20** can be reduced.

## Claims

1. A vehicle comprising: a vehicle body comprising a floor panel; a battery pack located below the floor panel; and an energy absorption member located outwardly from the battery pack in a vehicle width direction and fixing the battery pack to the vehicle body, wherein the energy absorption member is a hollow member extending along a vehicle longitudinal direction and having a constant cross section along the vehicle longitudinal direction, and comprises an inner sidewall that faces the battery pack, the inner sidewall of the energy absorption member inclines downwardly in a vehicle height direction as it extends inwardly in the vehicle width direction, the battery pack

comprises a bracket that protrudes outwardly in the vehicle width direction, the energy absorption member comprises an inner end portion in which the inner side wall is included and a middle portion adjacent to the inner end portion in the vehicle width direction, the inner end portion of the energy absorption member is fixed to a lower surface of the bracket of the battery pack from below, and an upper wall of the energy absorption member is located lower than the lower surface of the bracket in the vehicle height direction.

2. The vehicle according to claim 1, further comprising: a collar extending in the vehicle height direction and comprising an upper end being in contact with the battery pack and a lower end being in contact with the energy absorption member; and a bolt extending through the collar and fastening the energy absorption member to the battery pack.

3. The vehicle according to claim 2, wherein the inner sidewall of the energy absorption member comprises an opening through which the collar extends.

4. The vehicle according to claim 2, wherein the collar is welded to the inner sidewall of the energy absorption member.

5. The vehicle according to claim 2, wherein an inner diameter of the collar is at least twice a nominal diameter of the bolt.

6. The vehicle according to claim 2, wherein the bolt fastens the energy absorption member to the bracket of the battery pack.

7. The vehicle according to claim 1, wherein a thickness of the inner side wall of the energy absorption member is greater than a thickness of any one of a plurality of walls that form the middle portion of the energy absorption member.

8. The vehicle according to claim 7, wherein a thickness of any one of a plurality of walls that form the inner end portion is greater than the thickness of any one of a plurality of walls that form the middle portion of the energy absorption member.

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